SUPPLEMENTARY RESULTS

Modeling Localized Inflation in One-Truncated Count Data: An Application to Seabird Aggregation Sizes

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1. Purpose of this supplement

This QR-code supplement retains the results that add detail beyond the main poster. The poster remains the primary summary; the material below provides additional information on parameter uncertainty and computational reliability.

2. Parameter uncertainty

k	Median SE(p)	Median RSE(p)	Median SE(lambda)	Median RSE(lambda)
2	0.04365	0.122	0.3889	0.0365
10	0.00844	0.993	0.2458	0.0225
15	0.00523	0.748	0.2677	0.0136
20	0.00996	0.569	0.2368	0.0194
25	0.00308	0.705	0.2669	0.0109
30	0.00678	0.573	0.2590	0.0125

Relative uncertainty in p is lowest at k = 2. At larger k, the estimated inflation probability is often near zero, so RSE(p) becomes large even when its absolute standard error is small. In contrast, lambda is comparatively stable across the investigated inflation points.

This pattern is consistent with stronger identification of localized inflation at k = 2 and weaker relative identification of p when the fitted inflation component is small.

3. Computational and inferential reliability

k	Applicable fits	p-boundary	Identity check p=0	EM monotone
2	49	0	49 / 49 PASS	49 / 49
10	37	9	37 / 37 PASS	37 / 37
15	26	7	26 / 26 PASS	26 / 26
20	33	5	33 / 33 PASS	33 / 33
25	21	5	21 / 21 PASS	21 / 21
30	28	4	28 / 28 PASS	28 / 28

For every applicable OTIP fit, setting p = 0 reproduced the OTP identity, and the EM log-likelihood was monotone within numerical tolerance. A p-boundary result means that the estimated inflation probability is at or near its lower boundary, so OTIP is effectively behaving like OTP. This is statistical or identifiability behavior rather than an automatic optimization failure.

Interpretive note. Boundary behavior becomes more common at the larger investigated inflation points, where the data often provide less evidence for a distinct inflated component.

4. Supplementary takeaway

The additional uncertainty and diagnostic results support the poster's main interpretation: OTIP is most informative when localized point inflation is clearly present. The especially low median RSE for p at k = 2 is consistent with the strong empirical evidence for excess mass at group size 2. At larger k, p is more frequently near its boundary and relatively less precisely estimated, while lambda remains comparatively stable.

Prepared as a QR-code supplement to the 36 x 48 inch research poster. The model-selection and dispersion summaries already displayed on the poster have intentionally not been repeated here.